

class10: Structural Bioinformatics (pt1)

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Background

The main repository of high-resolution structural data on biomolecules is called the **Protein Data Bank** (PDB).

2. Introduction to the RCSB Protein Data Bank (PDB)

PDB statistics

What is in the PDB in terms of molecule type and structure determination method?

Read a CSV file of current PDB stats obtained from <https://www.rcsb.org/stats/summary>

```
pdb <- read.csv("Data Export Summary.csv") # "CSV" button on website
pdb
```

	Molecular.Type	X.ray	EM	NMR	Integrative	Multiple.methods
1	Protein (only)	180,758	23,111	12,813	348	229
2	Protein/Oligosaccharide	10,488	3,741	34	8	11
3	Protein/NA	9,205	6,751	287	26	8
4	Nucleic acid (only)	3,154	250	1,578	3	15
5	Other	178	27	35	4	0
6	Oligosaccharide (only)	11	0	6	0	1
	Neutron	Other	Total			
1	84	32	217,375			
2	1	0	14,283			
3	0	0	16,277			
4	3	1	5,004			
5	0	0	244			
6	0	4	22			

Q1.

What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy?

```
pdb$X.ray
```

```
[1] "180,758" "10,488" "9,205" "3,154" "178" "11"
```

This print out above `pdb$X.ray` is “character”, not “numeric”. Therefore, I can’t do math with it. We need to fix this..

1. Two functions that can help here are `sub()` and `as.numeric()`

```
#We want to get rid (or sub out) commas:
```

```
x <- pdb$X.ray
tmp <- sub(",", "", x)
sum( as.numeric(tmp))
```

```
[1] 203794
```

2. We could make a function to do this:

```
rm.comma <- function(x) {
  tmp <- sub(",", "", x)
  sum( as.numeric(tmp))
}
```

```
n.tot <- rm.comma(pdb$Total)
n.xray <- rm.comma(pdb$`X-ray`)
n.em <- rm.comma(pdb$EM)

n.xray/n.tot * 100
```

```
[1] 0
```

```
n.em/n.tot * 100
```

```
[1] 13.38046
```

3. We could also use a different import function for this CSV that speaks American (i.e. deals with commas in numbers in a comma separated value file)

```
library(readr) #CONSOLE: install.packages("readr")
```

Warning: package 'readr' was built under R version 4.5.2

```
pdb <- read_csv("Data Export Summary.csv")
```

Rows: 6 Columns: 9

-- Column specification -----

Delimiter: ","

chr (1): Molecular Type

dbl (4): Integrative, Multiple methods, Neutron, Other

num (4): X-ray, EM, NMR, Total

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
n.tot <- sum(pdb$Total)
```

```
n.xray <- sum(pdb$`X-ray`)
```

```
n.em <- sum(pdb$EM)
```

```
100 * n.xray / n.tot
```

[1] 80.48577

```
100 * n.em / n.tot
```

[1] 13.38046

80.4% are solved by X-Ray and 13.38% are solved by Electron Microscopy

Q. How many total protein structures are there in the dataset?

```
pdb$Total[1]
```

[1] 217375

217375 total protein structures in the dataset

Q2.

What proportion of structures in the PDB are protein?

The total number of protein sequences in UniProt is 202,556,314

```
217375/202556314 * 100
```

```
[1] 0.1073158
```

0.1073158% structures in the PDB are protein

Key-point: We have a very, very small structural coverage of known proteins (~0.1%). Most structures we know about (~80%) come from one method (X-ray crystallography)

Q3.

Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

There are currently 1,227 HIV-1 protease structures in the PDB.

3. Visualizing the HIV-1 protease structure

Visualizing PDB data with Mol-star (Using Mol*)

Main stand alone web version with all features is at <https://molstar.org/viewer/>

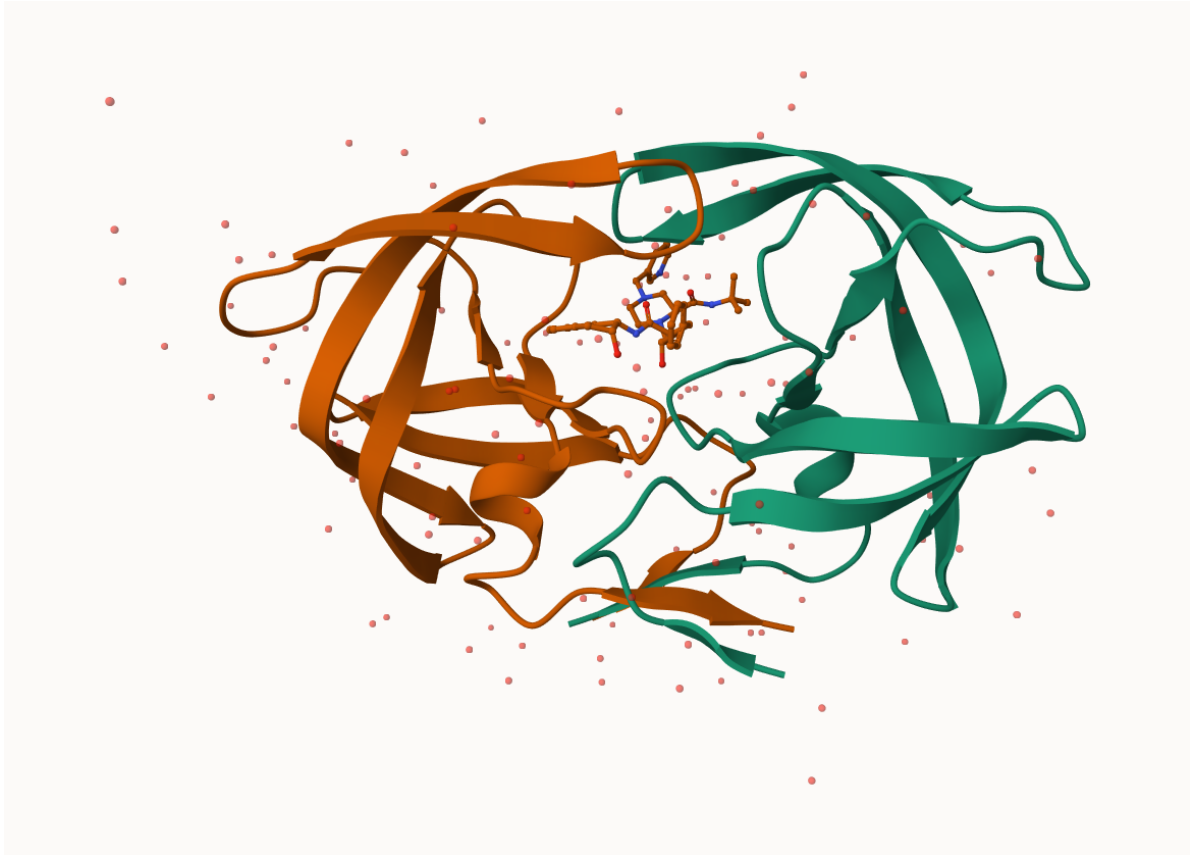


Figure 1: The HIV-protease enzyme is a homodimer with two chains

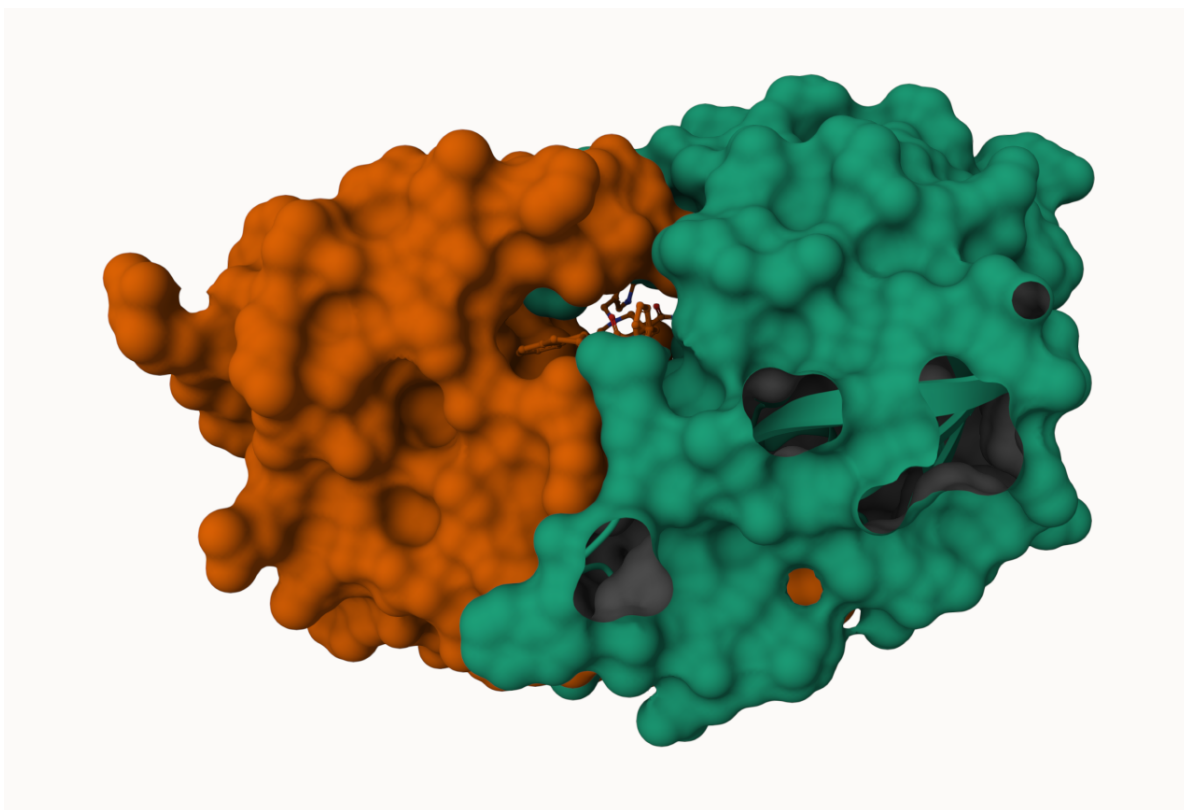


Figure 2: Surface display showing the binding cleft size for the inhibitor (drug) molecule

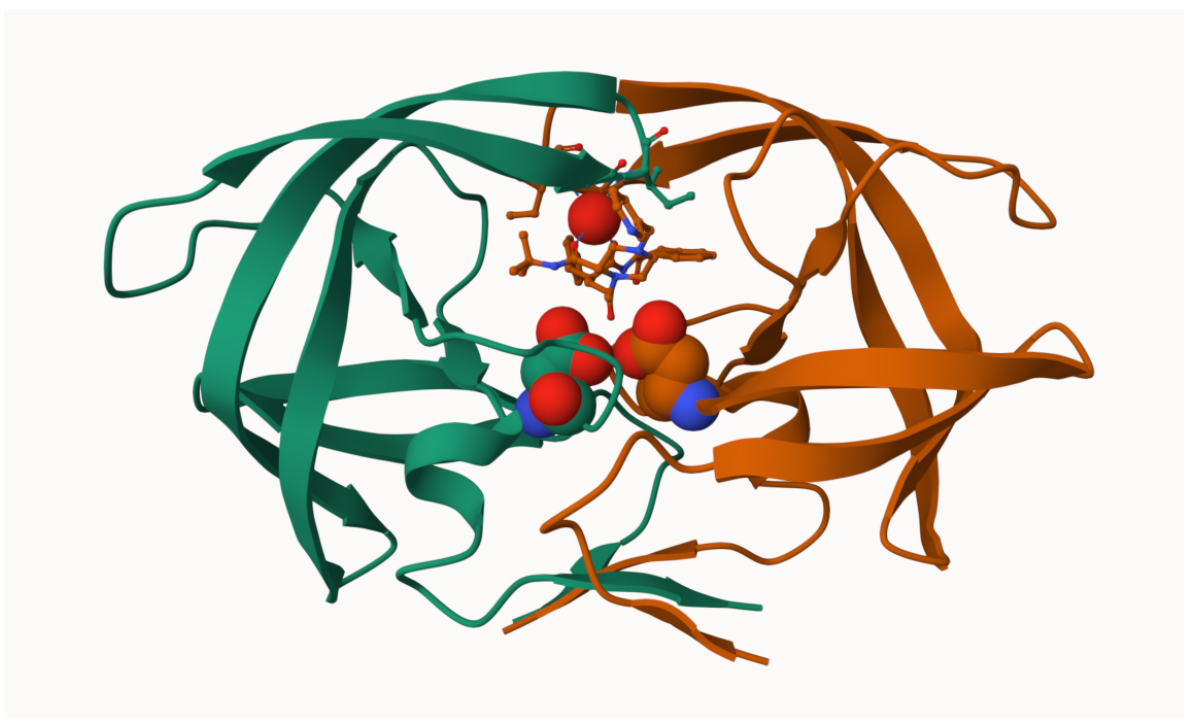


Figure 3: Spacefill display of catalytic ASP25 amino acids and key binding site water molecule

The important role of water

Q4.

Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

1HSG is an X-ray diffraction structure at 2.00 Å resolution. Hydrogen is a very small molecule and thus not resolved in X-ray data, in which one atom per water molecule is seen (represented by the oxygen atom).

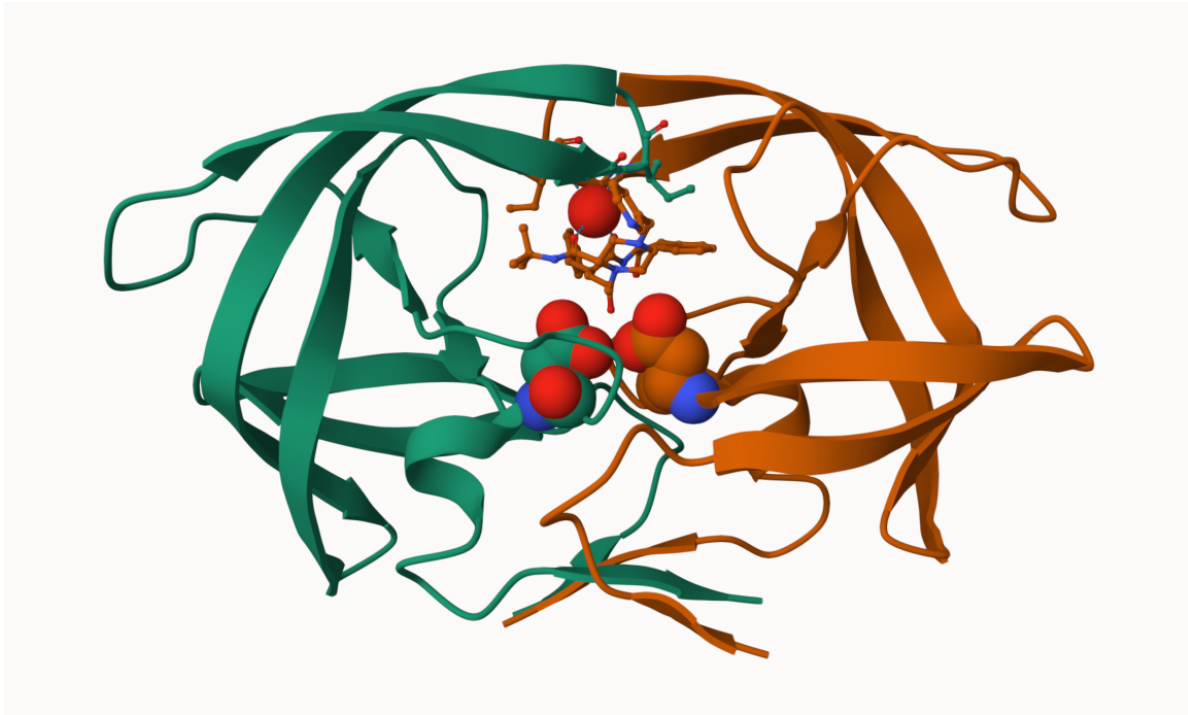
Q5.

There is a critical “conserved” water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have?

Yes, the water molecule is identified on Chain B. The water molecule’s residue number is HOH 308.

Q6.

Generate and save a figure clearly showing the two distinct chains of HIV-protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend “Ball & Stick” for these side-chains). Add this figure to your Quarto document.



The two multi-colored spacefill models represent the A and B chain of ASP25, and the red spacefill model above the ASP25 models is the critical water molecule.

Discussion Topic: Can you think of a way in which indinavir, or even larger ligands and substrates, could enter the binding site?

Indinavir, or even larger ligands and substrates, could enter the binding site through changes in the HIV protease where the two flap regions over the binding pocket could open and allow access to the exposed active site.

4. Introduction to Bio3D in R

Getting Started with the Bio3D package

Bio3D is an R package from CRAN for structural bioinformatics

```
library(bio3d) #CONSOLE: install.packages("bio3d")
pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

pdb

```
Call: read.pdb(file = "1hsg")
```

```
Total Models#: 1
```

```
Total Atoms#: 1686, XYZs#: 5058 Chains#: 2 (values: A B)
```

```
Protein Atoms#: 1514 (residues/Calpha atoms#: 198)
```

```
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)
```

```
Non-protein/nucleic Atoms#: 172 (residues: 128)
```

```
Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]
```

```
Protein sequence:
```

```
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
VNIIGRNLLTQIGCTLNF
```

```
+ attr: atom, xyz, seqres, helix, sheet,
      calpha, remark, call
```

Reading PDB file data into R

Q7.

How many amino acid residues are there in this pdb object?

198 amino acid residues.

Q8.

Name one of the two non-protein residues?

HOH (water), or MK1 (ligand).

Q9.

How many protein chains are in this structure?

2 protein chains are in this structure.

```
attributes(pdb)
```

```
$names
```

```
[1] "atom" "xyz" "seqres" "helix" "sheet" "calpha" "remark" "call"
```

```
$class
```

```
[1] "pdb" "sse"
```

```
head(pdb$atom) #access individual attributes
```

	type	eleno	elety	alt	resid	chain	resno	insert	x	y	z	o	b
1	ATOM	1	N	<NA>	PRO	A	1	<NA>	29.361	39.686	5.862	1	38.10
2	ATOM	2	CA	<NA>	PRO	A	1	<NA>	30.307	38.663	5.319	1	40.62
3	ATOM	3	C	<NA>	PRO	A	1	<NA>	29.760	38.071	4.022	1	42.64
4	ATOM	4	O	<NA>	PRO	A	1	<NA>	28.600	38.302	3.676	1	43.40
5	ATOM	5	CB	<NA>	PRO	A	1	<NA>	30.508	37.541	6.342	1	37.87
6	ATOM	6	CG	<NA>	PRO	A	1	<NA>	29.296	37.591	7.162	1	38.40

	segid	elesy	charge
1	<NA>	N	<NA>
2	<NA>	C	<NA>
3	<NA>	C	<NA>
4	<NA>	O	<NA>
5	<NA>	C	<NA>
6	<NA>	C	<NA>

There are lots of functions that can work with these `pdb` objects

```
head(pdbseq(pdb))
```

```
  1    2    3    4    5    6  
"P" "Q" "I" "T" "L" "W"
```

Quick PDB visualization in R

We can have a quick interactive view of any of these `pdb` objects:

```
library(bio3dview)
view.pdb(pdb)
```

Let's try a custom view

```
view.pdb(pdb,
  colorScheme = "sse",
  backgroundColor = "black")
```

Q. Create a custom view of HIV-Pr highlighting the active site ASP residues (resno=25), the two chains (in your choice of colors), and the ligand all on a custom color background?

```
library(bio3dview)
library(NGLViewerR)

active.site <- atom.select(pdb, resno=25)
view.pdb(pdb,
  cols = c("red", "blue"),
  highlight = active.site,
  highlight.style = "spacefill",
  backgroundColor = "lightpink") |>
  setRock()
```

Predicting functional motions of a single structure

Let's do a Normal Model Analysis (NMA) to predict the flexibility of a given `pdb` object:

```
adk <- read.pdb("6s36")
```

Note: Accessing on-line PDB file

PDB has ALT records, taking A only, `rm.alt=TRUE`

```
#A quick structure summary
adk
```

```
Call: read.pdb(file = "6s36")
```

```
Total Models#: 1
```

```
Total Atoms#: 1898, XYZs#: 5694 Chains#: 1 (values: A)
```

```
Protein Atoms#: 1654 (residues/Calpha atoms#: 214)
```

```
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)
```

```
Non-protein/nucleic Atoms#: 244 (residues: 244)
```

```
Non-protein/nucleic resid values: [ CL (3), HOH (238), MG (2), NA (1) ]
```

```
Protein sequence:
```

```
MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLV  
DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDKI  
VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQM  
TAPLIG  
YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
```

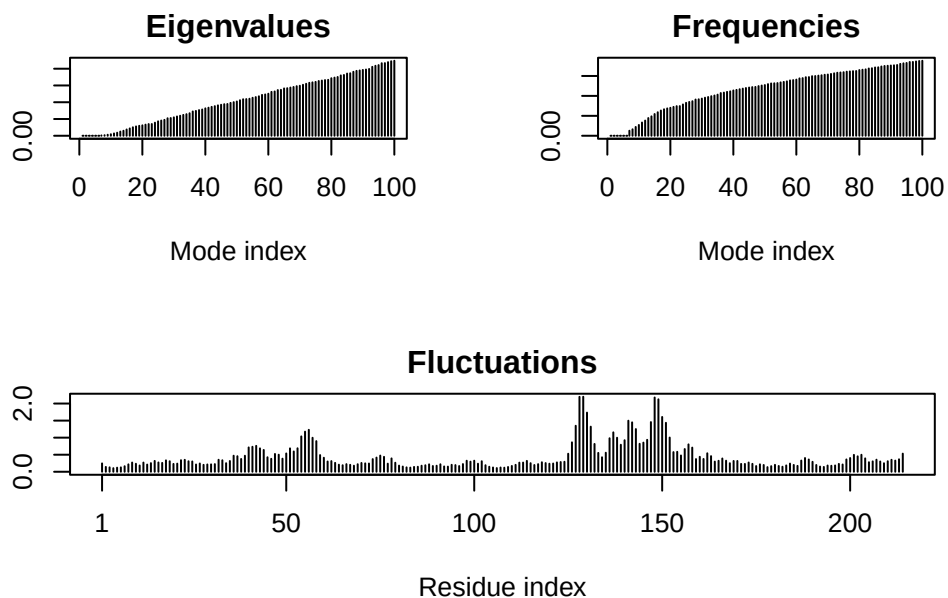
```
+ attr: atom, xyz, seqres, helix, sheet,  
      calpha, remark, call
```

```
# Perform flexibility prediction  
m <- nma(adk)
```

```
Building Hessian... Done in 0.01 seconds.
```

```
Diagonalizing Hessian... Done in 0.221 seconds.
```

```
plot(m)
```



View the results with an interactive structure view

```
view.nma(m)
```

Write out the results for viewing in Mol-star:

```
mktrj(m, file="adk_m7.pdb") #view a "movie" of predicted motions
# for quicker display:
view.nma(m, pdb=adk)
```

5. Comparative structure analysis of Adenylate Kinase (ADK) family

Setup

Install packages in console

Q10.

Which of the packages above is found only on BioConductor and not CRAN?

The msa package.

Q11.

Which of the above packages is not found on BioConductor or CRAN?:

The package “bio3dview” is not found on BioConductor or CRAN.

Q12.

True or False? Functions from the pak package can be used to install packages from GitHub and BitBucket?

TRUE

Search and retrieve ADK structures

Our first step is to find a sequence for this family. We will use the database ID “1ake-A” here:

```
library(bio3d)
id <- "1ake_A"

aa <- get.seq(id)
```

Warning in get.seq(id): Removing existing file: seqs.fasta

Fetching... Please wait. Done.

```
aa
```

```

      1      .      .      .      .      .      .      60
pdb|1AKE|A  MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLRAAVKSGSELGKQAKDIMDAGKLV
      1      .      .      .      .      .      .      60

      61      .      .      .      .      .      .      120
pdb|1AKE|A  DELVIALVKERIAQEDCRNGFLLDGFRTIPQADAMKEAGINVDYVLEFDVPDELIVDRI
      61      .      .      .      .      .      .      120

      121      .      .      .      .      .      .      180
pdb|1AKE|A  VGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
      121      .      .      .      .      .      .      180

      181      .      .      .      214
pdb|1AKE|A  YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
      181      .      .      .      214

```

Call:

```
read.fasta(file = outfile)
```

Class:

```
fasta
```

Alignment dimensions:

```
1 sequence rows; 214 position columns (214 non-gap, 0 gap)
```

```
+ attr: id, ali, call
```

Q13.

How many amino acids are in this sequence, i.e. how long is this sequence?

214 amino acids are in this sequence.

Search for related sequences in the database

```
blast <- blast.pdb(aa)
```

Searching ... please wait (updates every 5 seconds) RID = 1R9G8WKZ014

.....

Reporting 96 hits


```
head(blast$hit.tbl)
```

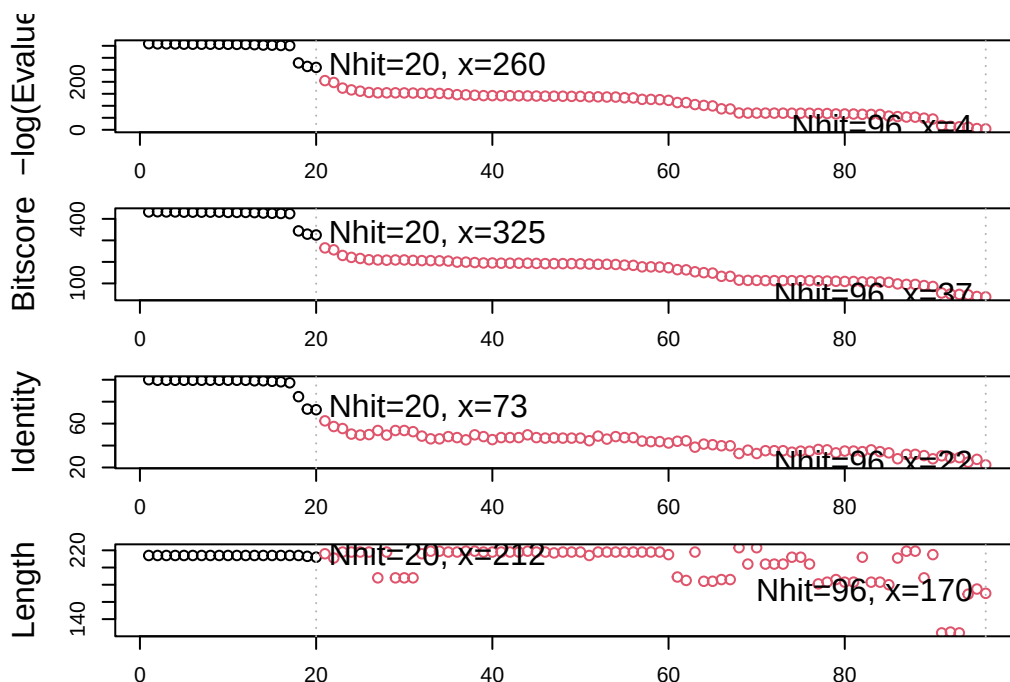
	queryid	subjectids	identity	alignmentlength	mismatches	gapopens	q.start
1	Query_6082395	1AKE_A	100.000	214	0	0	1
2	Query_6082395	8BQF_A	99.533	214	1	0	1
3	Query_6082395	4X8M_A	99.533	214	1	0	1
4	Query_6082395	6S36_A	99.533	214	1	0	1
5	Query_6082395	9R6U_A	99.533	214	1	0	1
6	Query_6082395	9R71_A	99.533	214	1	0	1

	q.end	s.start	s.end	eval	bitscore	positives	mlog.eval	pdb.id	acc
1	214	1	214	1.82e-156	432	100.00	358.6044	1AKE_A	1AKE_A
2	214	21	234	2.98e-156	433	100.00	358.1114	8BQF_A	8BQF_A
3	214	1	214	3.26e-156	432	100.00	358.0215	4X8M_A	4X8M_A
4	214	1	214	4.78e-156	432	100.00	357.6388	6S36_A	6S36_A
5	214	1	214	1.07e-155	431	99.53	356.8330	9R6U_A	9R6U_A
6	214	1	214	1.26e-155	431	99.53	356.6696	9R71_A	9R71_A

```
hits <- plot(blast)
```

```
* Possible cutoff values: 260 3
    Yielding Nhits: 20 96

* Chosen cutoff value of: 260
    Yielding Nhits: 20
```



```
hits$pdb.id
```

```
[1] "1AKE_A" "8BQF_A" "4X8M_A" "6S36_A" "9R6U_A" "9R71_A" "8Q2B_A" "8RJ9_A"
[9] "6RZE_A" "4X8H_A" "3HPR_A" "1E4V_A" "5EJE_A" "1E4Y_A" "3X2S_A" "6HAP_A"
[17] "6HAM_A" "8PVW_A" "4K46_A" "4NP6_A"
```

```
files <- get.pdb(hits$pdb.id, path="pdbc", split=TRUE, gzip=TRUE)
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbc", split = TRUE, gzip = TRUE):
pdbc/1AKE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbc", split = TRUE, gzip = TRUE):
pdbc/8BQF.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbc", split = TRUE, gzip = TRUE):
pdbc/4X8M.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbc", split = TRUE, gzip = TRUE):
pdbc/6S36.pdb.gz exists. Skipping download
```

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/9R6U.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/9R71.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8Q2B.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8RJ9.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6RZE.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4X8H.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3HPR.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4V.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/5EJE.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4Y.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3X2S.pdb.gz exists. Skipping download

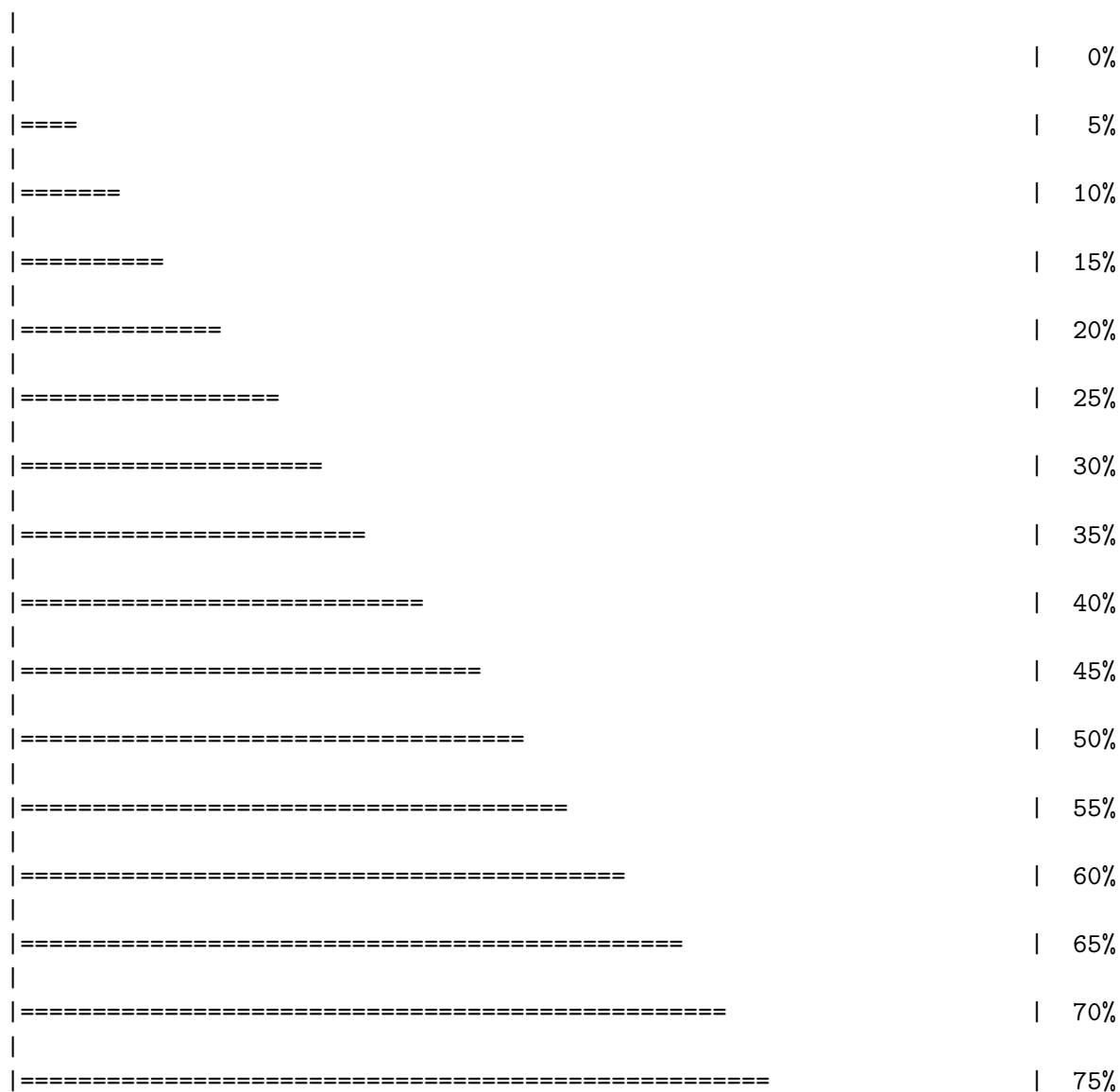
Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAP.pdb.gz exists. Skipping download

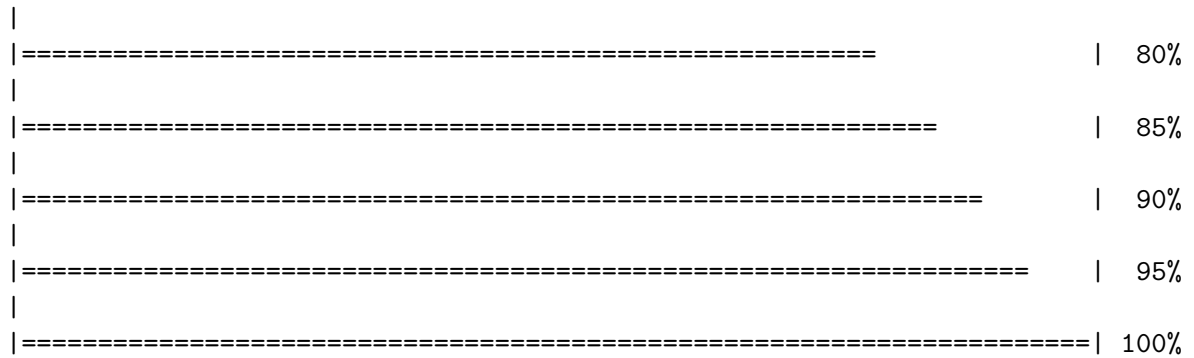
Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAM.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8PVW.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4K46.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4NP6.pdb.gz exists. Skipping download





Align and superpose structures

```
pdbbs <- pdbaln(files, fit = TRUE, exefile="msa")
```

Reading PDB files:

```
pdbbs/split_chain/1AKE_A.pdb
pdbbs/split_chain/8BQF_A.pdb
pdbbs/split_chain/4X8M_A.pdb
pdbbs/split_chain/6S36_A.pdb
pdbbs/split_chain/9R6U_A.pdb
pdbbs/split_chain/9R71_A.pdb
pdbbs/split_chain/8Q2B_A.pdb
pdbbs/split_chain/8RJ9_A.pdb
pdbbs/split_chain/6RZE_A.pdb
pdbbs/split_chain/4X8H_A.pdb
pdbbs/split_chain/3HPR_A.pdb
pdbbs/split_chain/1E4V_A.pdb
pdbbs/split_chain/5EJE_A.pdb
pdbbs/split_chain/1E4Y_A.pdb
pdbbs/split_chain/3X2S_A.pdb
pdbbs/split_chain/6HAP_A.pdb
pdbbs/split_chain/6HAM_A.pdb
pdbbs/split_chain/8PVW_A.pdb
pdbbs/split_chain/4K46_A.pdb
pdbbs/split_chain/4NP6_A.pdb
```

```
  PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
```

```

.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
.... PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..

```

Extracting sequences

```

pdb/seq: 1   name: pdbs/split_chain/1AKE_A.pdb
            PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 2   name: pdbs/split_chain/8BQF_A.pdb
            PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 3   name: pdbs/split_chain/4X8M_A.pdb
pdb/seq: 4   name: pdbs/split_chain/6S36_A.pdb
            PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 5   name: pdbs/split_chain/9R6U_A.pdb
            PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 6   name: pdbs/split_chain/9R71_A.pdb
            PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 7   name: pdbs/split_chain/8Q2B_A.pdb
            PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 8   name: pdbs/split_chain/8RJ9_A.pdb
            PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 9   name: pdbs/split_chain/6RZE_A.pdb
            PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 10  name: pdbs/split_chain/4X8H_A.pdb
pdb/seq: 11  name: pdbs/split_chain/3HPR_A.pdb
            PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 12  name: pdbs/split_chain/1E4V_A.pdb
pdb/seq: 13  name: pdbs/split_chain/5EJE_A.pdb
            PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 14  name: pdbs/split_chain/1E4Y_A.pdb
pdb/seq: 15  name: pdbs/split_chain/3X2S_A.pdb
pdb/seq: 16  name: pdbs/split_chain/6HAP_A.pdb
pdb/seq: 17  name: pdbs/split_chain/6HAM_A.pdb
            PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 18  name: pdbs/split_chain/8PVW_A.pdb
            PDB has ALT records, taking A only, rm.alt=TRUE

```

pdb/seq: 19 name: pdbs/split_chain/4K46_A.pdb
 PDB has ALT records, taking A only, rm.alt=TRUE
 pdb/seq: 20 name: pdbs/split_chain/4NP6_A.pdb

pdbs

	1	.	.	.	40
[Truncated_Name:1] 1AKE_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	M	L	R	A
[Truncated_Name:2] 8BQF_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	M	L	R	A
[Truncated_Name:3] 4X8M_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	M	L	R	A
[Truncated_Name:4] 6S36_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	M	L	R	A
[Truncated_Name:5] 9R6U_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	M	L	R	A
[Truncated_Name:6] 9R71_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	M	L	R	A
[Truncated_Name:7] 8Q2B_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	M	L	R	A
[Truncated_Name:8] 8RJ9_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	M	L	R	A
[Truncated_Name:9] 6RZE_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	M	L	R	A
[Truncated_Name:10] 4X8H_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	M	L	R	A
[Truncated_Name:11] 3HPR_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	M	L	R	A
[Truncated_Name:12] 1E4V_A.pdb	--MRIILLGAPVAGKGTQAQFIMEKYGIPQISTG	M	L	R	A
[Truncated_Name:13] 5EJE_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	M	L	R	A
[Truncated_Name:14] 1E4Y_A.pdb	--MRIILLGALVAGKGTQAQFIMEKYGIPQISTG	M	L	R	A
[Truncated_Name:15] 3X2S_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	M	L	R	A
[Truncated_Name:16] 6HAP_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	M	L	R	A
[Truncated_Name:17] 6HAM_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	M	L	R	A
[Truncated_Name:18] 8PVW_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	M	L	R	A
[Truncated_Name:19] 4K46_A.pdb	--MRIILLGAPGAGKGTQAQFIMAKFGIPQISTG	M	L	R	A
[Truncated_Name:20] 4NP6_A.pdb	NAMRIILLGAPGAGKGTQAQFIMEKFGIPQISTG	M	L	R	A
	***** ***** *~*****				
	1	.	.	.	40
	41	.	.	.	80
[Truncated_Name:1] 1AKE_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:2] 8BQF_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQE---				
[Truncated_Name:3] 4X8M_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:4] 6S36_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:5] 9R6U_A.pdb	VKSGSELGAQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:6] 9R71_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:7] 8Q2B_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:8] 8RJ9_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:9] 6RZE_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:10] 4X8H_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:11] 3HPR_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				

[Truncated_Name:7] 8Q2B_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKA
[Truncated_Name:8] 8RJ9_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:9] 6RZE_A.pdb	AIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:10] 4X8H_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:11] 3HPR_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDGTGEELTTRKD
[Truncated_Name:12] 1E4V_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:13] 5EJE_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:14] 1E4Y_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:15] 3X2S_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:16] 6HAP_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:17] 6HAM_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:18] 8PVW_A.pdb	RILKR--GETSGRV-----D
[Truncated_Name:19] 4K46_A.pdb	RMAGRRAHLASGRTYHNVNPPKVEGKDDVTGEDLVIRE
[Truncated_Name:20] 4NP6_A.pdb	RMAGRRAHLPSGRTYHVYNPPKVEGKDDVTGEDLVIRE
	^ * ***
	121 . . . 160
	161 . . . 200
[Truncated_Name:1] 1AKE_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:2] 8BQF_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:3] 4X8M_A.pdb	DQEETVRKRLVEWHQM TAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:4] 6S36_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:5] 9R6U_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:6] 9R71_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:7] 8Q2B_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:8] 8RJ9_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:9] 6RZE_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:10] 4X8H_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:11] 3HPR_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:12] 1E4V_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:13] 5EJE_A.pdb	DQEECVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:14] 1E4Y_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:15] 3X2S_A.pdb	DQEETVRKRLCEYHQM TAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:16] 6HAP_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:17] 6HAM_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:18] 8PVW_A.pdb	DNEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:19] 4K46_A.pdb	DKEETVLARLGVYHNQTAPLIAYYGKEAEAGNTQYLKFDG
[Truncated_Name:20] 4NP6_A.pdb	DKEETVRARLNVYHTQTAPLIEYYGKEAAAGKTQYLKFDG
	* * * * * ^ * * * * * * * * * *
	161 . . . 200
	201 . 216
[Truncated_Name:1] 1AKE_A.pdb	TKPVAEVRADLEKILG

```

[Truncated_Name:2]8BQF_A.pdb      TKPVAEVRADLEKIL-
[Truncated_Name:3]4X8M_A.pdb      TKPVAEVRADLEKILG
[Truncated_Name:4]6S36_A.pdb      TKPVAEVRADLEKILG
[Truncated_Name:5]9R6U_A.pdb      TKPVAEVRADLEKILG
[Truncated_Name:6]9R71_A.pdb      TKPVAEVRADLEKILG
[Truncated_Name:7]8Q2B_A.pdb      TKPVAEVRADLEKILG
[Truncated_Name:8]8RJ9_A.pdb      TKPVAEVRADLEKILG
[Truncated_Name:9]6RZE_A.pdb      TKPVAEVRADLEKILG
[Truncated_Name:10]4X8H_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:11]3HPR_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:12]1E4V_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:13]5EJE_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:14]1E4Y_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:15]3X2S_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:16]6HAP_A.pdb     TKPVCEVRADLEKILG
[Truncated_Name:17]6HAM_A.pdb     TKPVCEVRADLEKILG
[Truncated_Name:18]8PVW_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:19]4K46_A.pdb     TKAVAEVSAELEKALA
[Truncated_Name:20]4NP6_A.pdb     TKQVSEVSADIAKALA
                                   ** * ** *~* *
                                   201      .      216

```

Call:

```
pdbaln(files = files, fit = TRUE, exefile = "msa")
```

Class:

```
pdbs, fasta
```

Alignment dimensions:

```
20 sequence rows; 216 position columns (182 non-gap, 34 gap)
```

```
+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

Quick interactive structural view

```
library(bio3dview)
view.pdbs(pdbs, colorScheme = "residueIndex")
```

Annotate collected PDB structures

```
# Vector containing PDB database codes
ids <- basename.pdb(pdb$ids)

anno <- pdb.annotate(ids)
unique(anno$source)
```

```
[1] "Escherichia coli"
[2] "Escherichia coli K-12"
[3] "Escherichia coli O139:H28 str. E24377A"
[4] "Escherichia coli str. K-12 substr. MDS42"
[5] "Photobacterium profundum"
[6] "Vibrio cholerae O1 biovar El Tor str. N16961"
```

```
anno
```

	structureId	chainId	macromoleculeType	chainLength	experimentalTechnique
1AKE_A	1AKE	A	Protein	214	X-ray
8BQF_A	8BQF	A	Protein	234	X-ray
4X8M_A	4X8M	A	Protein	214	X-ray
6S36_A	6S36	A	Protein	214	X-ray
9R6U_A	9R6U	A	Protein	214	X-ray
9R71_A	9R71	A	Protein	214	X-ray
8Q2B_A	8Q2B	A	Protein	214	X-ray
8RJ9_A	8RJ9	A	Protein	214	X-ray
6RZE_A	6RZE	A	Protein	214	X-ray
4X8H_A	4X8H	A	Protein	214	X-ray
3HPR_A	3HPR	A	Protein	214	X-ray
1E4V_A	1E4V	A	Protein	214	X-ray
5EJE_A	5EJE	A	Protein	214	X-ray
1E4Y_A	1E4Y	A	Protein	214	X-ray
3X2S_A	3X2S	A	Protein	214	X-ray
6HAP_A	6HAP	A	Protein	214	X-ray
6HAM_A	6HAM	A	Protein	214	X-ray
8PVW_A	8PVW	A	Protein	187	X-ray
4K46_A	4K46	A	Protein	214	X-ray
4NP6_A	4NP6	A	Protein	217	X-ray
	resolution	scopDomain	pfam		
1AKE_A	2.000	Adenylate kinase	Adenylate kinase, active site lid (ADK_lid)		
8BQF_A	2.050	<NA>	Adenylate kinase, active site lid (ADK_lid)		
4X8M_A	2.600	<NA>	Adenylate kinase, active site lid (ADK_lid)		
6S36_A	1.600	<NA>	Adenylate kinase, active site lid (ADK_lid)		

9R6U_A	1.770	<NA>	Adenylate kinase, active site lid (ADK_lid)
9R71_A	1.610	<NA>	Adenylate kinase, active site lid (ADK_lid)
8Q2B_A	1.760	<NA>	Adenylate kinase (ADK)
8RJ9_A	1.590	<NA>	Adenylate kinase, active site lid (ADK_lid)
6RZE_A	1.690	<NA>	Adenylate kinase (ADK)
4X8H_A	2.500	<NA>	Adenylate kinase (ADK)
3HPR_A	2.000	<NA>	Adenylate kinase, active site lid (ADK_lid)
1E4V_A	1.850	Adenylate kinase	Adenylate kinase (ADK)
5EJE_A	1.900	<NA>	Adenylate kinase, active site lid (ADK_lid)
1E4Y_A	1.850	Adenylate kinase	Adenylate kinase, active site lid (ADK_lid)
3X2S_A	2.800	<NA>	<NA>
6HAP_A	2.700	<NA>	Adenylate kinase, active site lid (ADK_lid)
6HAM_A	2.550	<NA>	Adenylate kinase, active site lid (ADK_lid)
8PVW_A	2.000	<NA>	Adenylate kinase, active site lid (ADK_lid)
4K46_A	2.010	<NA>	Adenylate kinase, active site lid (ADK_lid)
4NP6_A	2.004	<NA>	Adenylate kinase, active site lid (ADK_lid)

ligandId

1AKE_A	AP5
8BQF_A	AP5
4X8M_A	<NA>
6S36_A	CL (3),NA,MG (2)
9R6U_A	AP5,GOL,NA
9R71_A	AP5
8Q2B_A	AP5,S04,MPO
8RJ9_A	ADP (2)
6RZE_A	NA (3),CL (2)
4X8H_A	<NA>
3HPR_A	AP5
1E4V_A	AP5
5EJE_A	AP5,CO
1E4Y_A	AP5
3X2S_A	JPY (2),AP5,MG
6HAP_A	AP5
6HAM_A	AP5
8PVW_A	AP5
4K46_A	ADP,AMP,PO4
4NP6_A	<NA>

ligandName

1AKE_A	BIS(ADENOSINE)-5'-PENTAPHOSPHATE
8BQF_A	BIS(ADENOSINE)-5'-PENTAPHOSPHATE
4X8M_A	<NA>
6S36_A	CHLORIDE ION (3),SODIUM ION,MAGNESIUM ION (2)
9R6U_A	BIS(ADENOSINE)-5'-PENTAPHOSPHATE,GLYCEROL,SODIUM ION

9R71_A		BIS(ADENOSINE)-5'-PENTAPHOSPHATE
8Q2B_A	BIS(ADENOSINE)-5'-PENTAPHOSPHATE,SULFATE ION,3[N-MORPHOLINO]PROPANE SULFONIC ACID	
8RJ9_A		ADENOSINE-5'-DIPHOSPHATE (2)
6RZE_A		SODIUM ION (3),CHLORIDE ION (2)
4X8H_A		<NA>
3HPR_A		BIS(ADENOSINE)-5'-PENTAPHOSPHATE
1E4V_A		BIS(ADENOSINE)-5'-PENTAPHOSPHATE
5EJE_A	BIS(ADENOSINE)-5'-PENTAPHOSPHATE,COBALT (II) ION	
1E4Y_A		BIS(ADENOSINE)-5'-PENTAPHOSPHATE
3X2S_A	N-(pyren-1-ylmethyl)acetamide (2),BIS(ADENOSINE)-5'-PENTAPHOSPHATE,MAGNESIUM ION	
6HAP_A		BIS(ADENOSINE)-5'-PENTAPHOSPHATE
6HAM_A		BIS(ADENOSINE)-5'-PENTAPHOSPHATE
8PVW_A		BIS(ADENOSINE)-5'-PENTAPHOSPHATE
4K46_A	ADENOSINE-5'-DIPHOSPHATE,ADENOSINE MONOPHOSPHATE,PHOSPHATE ION	
4NP6_A		<NA>

	source
1AKE_A	Escherichia coli
8BQF_A	Escherichia coli
4X8M_A	Escherichia coli
6S36_A	Escherichia coli
9R6U_A	Escherichia coli
9R71_A	Escherichia coli
8Q2B_A	Escherichia coli
8RJ9_A	Escherichia coli
6RZE_A	Escherichia coli
4X8H_A	Escherichia coli
3HPR_A	Escherichia coli K-12
1E4V_A	Escherichia coli
5EJE_A	Escherichia coli 0139:H28 str. E24377A
1E4Y_A	Escherichia coli
3X2S_A	Escherichia coli str. K-12 substr. MDS42
6HAP_A	Escherichia coli 0139:H28 str. E24377A
6HAM_A	Escherichia coli K-12
8PVW_A	Escherichia coli K-12
4K46_A	Photobacterium profundum
4NP6_A	Vibrio cholerae 01 biovar El Tor str. N16961

1AKE_A	STRUCTURE OF THE COMPLEX BETWEEN ADENYLATE KINASE FROM ESCHERICHIA COLI AND THE INHIBIT
8BQF_A	
4X8M_A	
6S36_A	
9R6U_A	
9R71_A	

C
Cry

8Q2B_A	E. coli Adenylate Kinase variant D158A (
8RJ9_A	E. coli adenylate kinase Asp84
6RZE_A	
4X8H_A	
3HPR_A	
1E4V_A	
5EJE_A	Crys
1E4Y_A	
3X2S_A	
6HAP_A	
6HAM_A	
8PVW_A	
4K46_A	
4NP6_A	

		citation	rObserved	rFree
1AKE_A	Muller, C.W., et al. J Mol Biology (1992)	0.19600	NA	
8BQF_A	Scheerer, D., et al. Proc Natl Acad Sci U S A (2023)	0.22073	0.25789	
4X8M_A	Kovermann, M., et al. Nat Commun (2015)	0.24910	0.30890	
6S36_A	Rogne, P., et al. Biochemistry (2019)	0.16320	0.23560	
9R6U_A	Mattsson, J., et al. Biochemistry (2025)	NA	0.22790	
9R71_A	Mattsson, J., et al. Biochemistry (2025)	0.19600	0.24400	
8Q2B_A	Nam, K., et al. J Chem Inf Model (2024)	0.18320	0.22440	
8RJ9_A	Nam, K., et al. Sci Adv (2024)	0.15190	0.20290	
6RZE_A	Rogne, P., et al. Biochemistry (2019)	0.18650	0.23500	
4X8H_A	Kovermann, M., et al. Nat Commun (2015)	0.19610	0.28950	
3HPR_A	Schrank, T.P., et al. Proc Natl Acad Sci U S A (2009)	0.21000	0.24320	
1E4V_A	Muller, C.W., et al. Proteins (1993)	0.19600	NA	
5EJE_A	Kovermann, M., et al. Proc Natl Acad Sci U S A (2017)	0.18890	0.23580	
1E4Y_A	Muller, C.W., et al. Proteins (1993)	0.17800	NA	
3X2S_A	Fujii, A., et al. Bioconjug Chem (2015)	0.20700	0.25600	
6HAP_A	Kantaev, R., et al. J Phys Chem B (2018)	0.22630	0.27760	
6HAM_A	Kantaev, R., et al. J Phys Chem B (2018)	0.20511	0.24325	
8PVW_A	Rodriguez, J.A., et al. To be published	0.18590	0.23440	
4K46_A	Cho, Y.-J., et al. To be published	0.17000	0.22290	
4NP6_A	Kim, Y., et al. To be published	0.18800	0.22200	

	rWork	spaceGroup
1AKE_A	0.19600	P 21 2 21
8BQF_A	0.21882	P 2 21 21
4X8M_A	0.24630	C 1 2 1
6S36_A	0.15940	C 1 2 1
9R6U_A	0.19190	P 21 2 21
9R71_A	0.19300	P 21 21 2
8Q2B_A	0.18100	P 1 21 1

```

8RJ9_A 0.15010 P 21 21 2
6RZE_A 0.18190 C 1 2 1
4X8H_A 0.19140 C 1 2 1
3HPR_A 0.20620 P 21 21 2
1E4V_A 0.19600 P 21 2 21
5EJE_A 0.18630 P 21 2 21
1E4Y_A 0.17800 P 1 21 1
3X2S_A 0.20700 P 21 21 21
6HAP_A 0.22370 I 2 2 2
6HAM_A 0.20311 P 43
8PVW_A 0.18340 P 2 21 21
4K46_A 0.16730 P 21 21 21
4NP6_A 0.18600 P 43

```

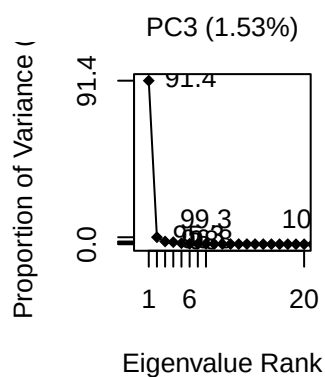
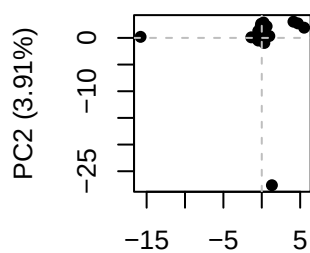
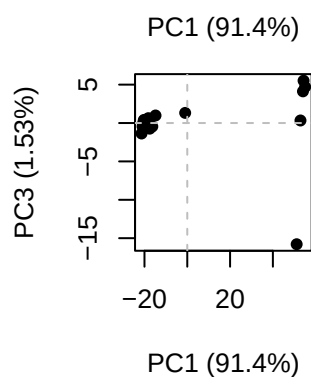
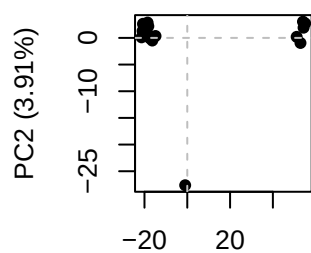
Principal component analysis

PCA of all this structural data (x,y and z atom coordinates)

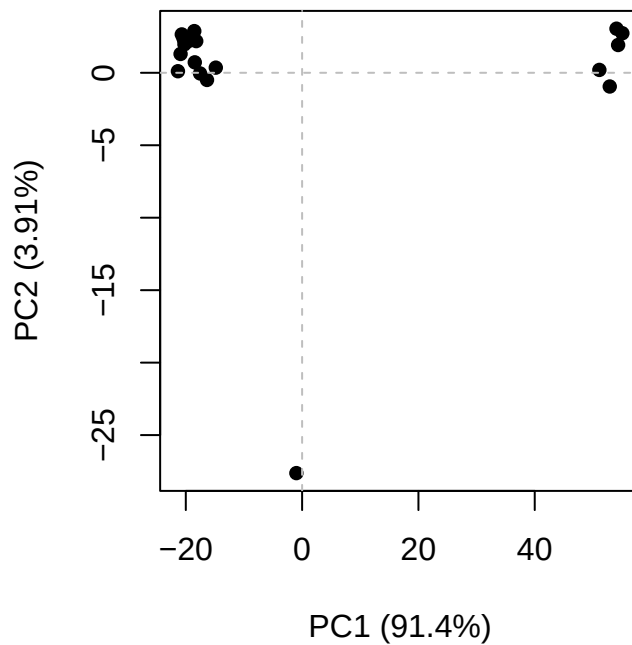
```

pc <- pca(pdfs)
plot(pc)

```



```
plot(pc, 1:2)
```

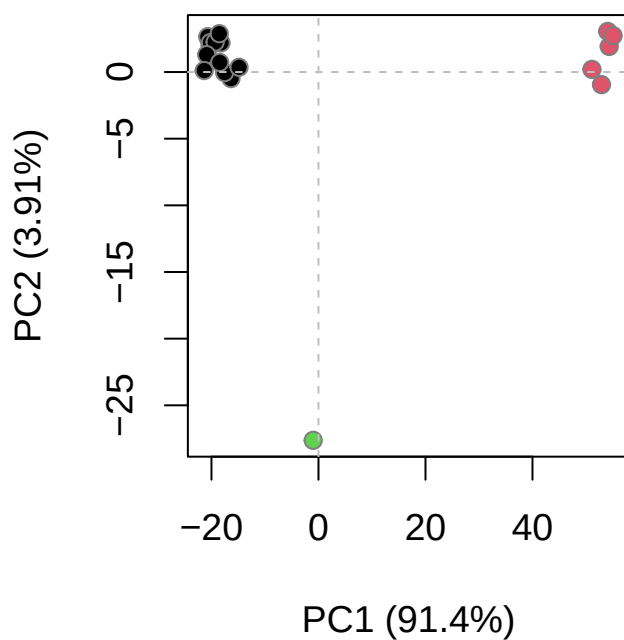


Function `rmsd()` facilitates clustering analysis based on pairwise structural deviation

```
# Calculate RMSD  
rd <- rmsd(pdb)
```

Warning in `rmsd(pdb)`: No indices provided, using the 182 non NA positions

```
# Structure-based clustering  
hc.rd <- hclust(dist(rd))  
grps.rd <- cutree(hc.rd, k=3)  
  
plot(pc, 1:2, col="grey50", bg=grps.rd, pch=21, cex=1)
```

PCA visualization

```
# Visualize first principal component
pc1 <- mktrj(pc, pc=1, file="pc_1.pdb")
pc1
```

```
Total Frames#: 34
Total XYZs#: 546, (Atoms#: 182)
```

```
[1] 26.417 52.833 39.777 <...> 16.853 51.184 40.052 [18564]
```

```
+ attr: Matrix DIM = 34 x 546
```

Interactive view of the PC1 captured structural differences:

```
view.pca(pc)
```

```
mktrj(pc, file = "pca.pdb")
```

We can also plot our main PCA results with ggplot:

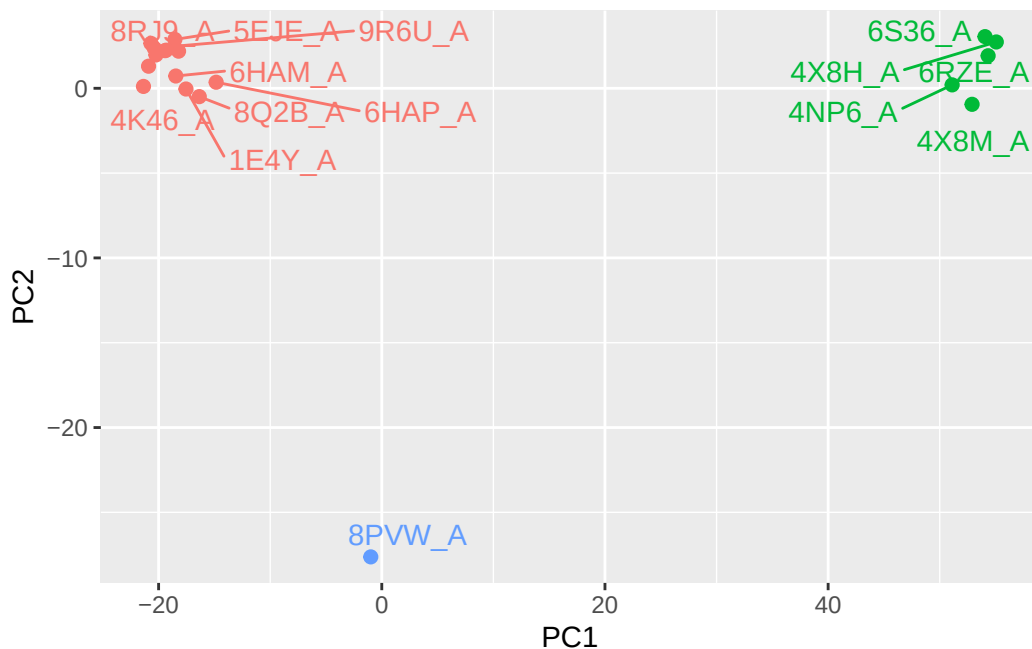
```
#Plotting results with ggplot2  
library(ggplot2)
```

Warning: package 'ggplot2' was built under R version 4.5.2

```
library(ggrepel)
```

Warning: package 'ggrepel' was built under R version 4.5.2

```
df <- data.frame(PC1=pc$z[,1],  
                 PC2=pc$z[,2],  
                 col=as.factor(grps.rd),  
                 ids=ids)  
  
p <- ggplot(df) +  
  aes(PC1, PC2, col=col, label=ids) +  
  geom_point(size=2) +  
  geom_text_repel(max.overlaps = 20) +  
  theme(legend.position = "none")  
p
```



6. Normal mode analysis [optional]

```
modes <- nma(pdbbs)
```

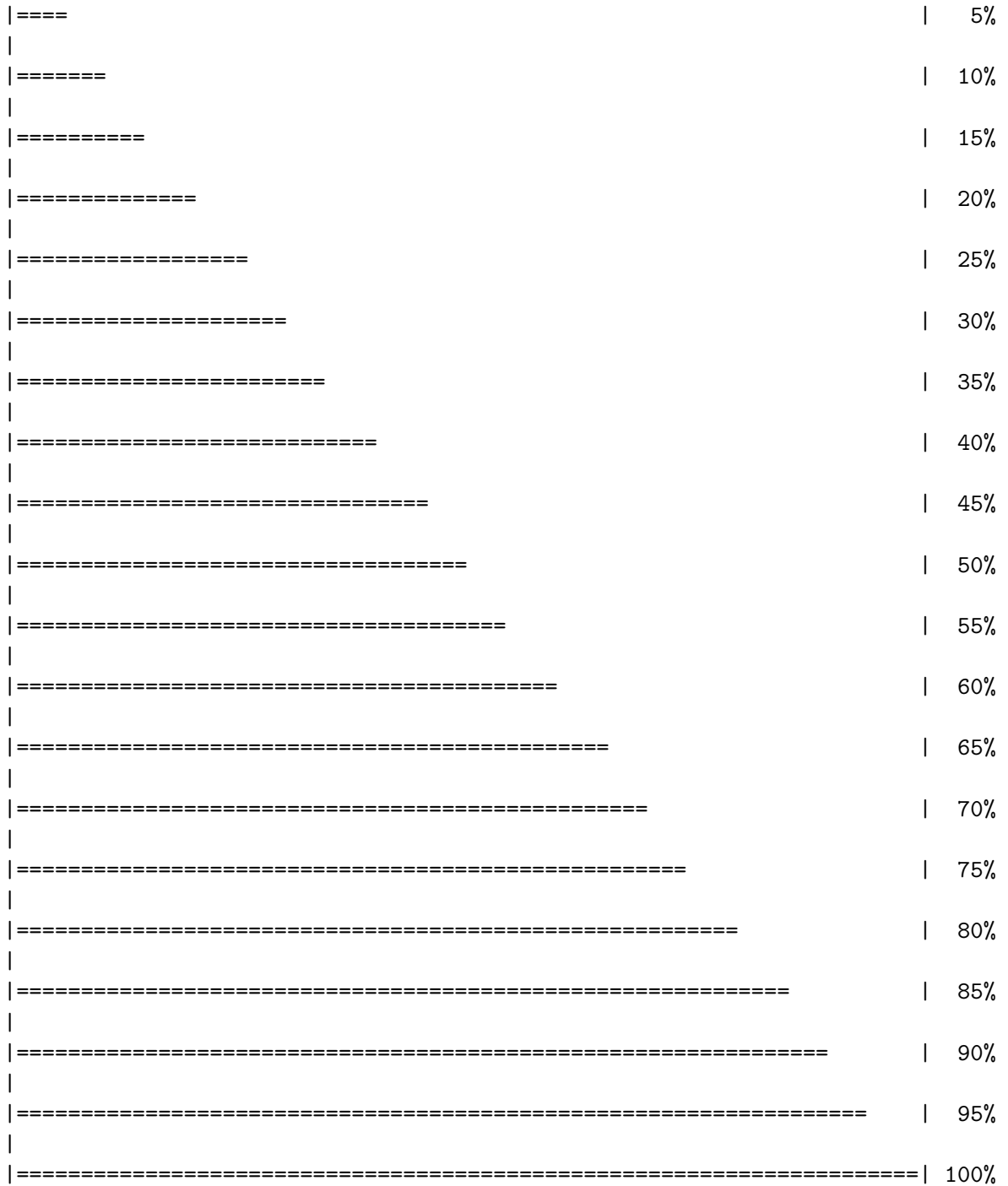
Warning in nma.pdbbs(pdbbs): 8BQF_A.pdb might have missing residue(s) in structure:
Fluctuations at neighboring positions may be affected.

Details of Scheduled Calculation:

```
... 20 input structures
... storing 540 eigenvectors for each structure
... dimension of x$U.subspace: ( 546x540x20 )
... coordinate superposition prior to NM calculation
... aligned eigenvectors (gap containing positions removed)
... estimated memory usage of final 'eNMA' object: 45.1 Mb
```

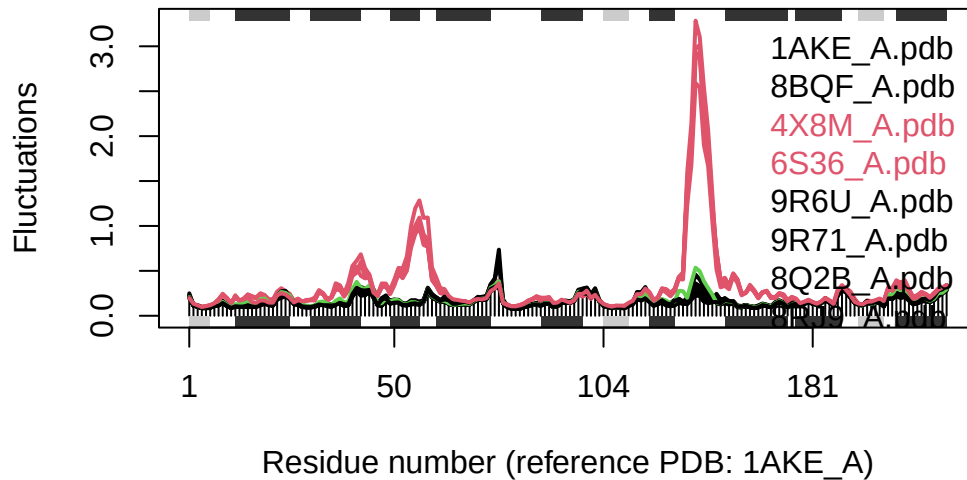
|
|
|

| 0%



```
plot(modes, pdb, col=grps.rd)
```

Extracting SSE from pdb\$sse attribute



Q14.

What do you note about this plot? Are the black and colored lines similar or different? Where do you think they differ most and why?

The black lines are lower, meaning that they have smaller fluctuations and likely represent a more stable, closed, or core-like conformation of the protein. The colored lines are showing greater fluctuations, especially around residues ~50 and ~150, suggesting that these regions are more flexible and likely correspond to mobile loop, hinge or domain motions.